EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 22- 300- 2
LED TUBE LIGHT T8-9W-60 cm	January 2026

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TECHNICAL SPECIFICATION

FOR

LED TUBE LIGHT

(T8−9W−60 cm) (MILKY)

Issue: Jan -2026/ Rev- 2

توجد بنود إختيارية Option items يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

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1 - Scope:

This specification covers the minimum technical requirements for design, engineering, manufacturing, testing and packing of **LED** tubes of T8-9W, 60 cm (Milky).

2 - Applicable standards:

Unless otherwise specified in this specification, the **LED** tubes of T8-9W, 60 cm (Milky) Should be designed, manufactured and tested according to the requirements of this specification, taking into consideration the recommendations and requirements of the latest editions of the following standards:

IEC 60081	Double-capped fluorescent lamps – Performance specifications
IEC 60598-1	Luminaires – Part 1: General requirements and tests
IEC 60598-2	Corrigendum 1 - Luminaires - Part 2: Particular requirements
IEC 62504	General lighting - Light emitting diode (LED)
IEC 62560	Corrigendum 2 - Self-ballasted LED-lamps for general lighting services
IEC 61347-2-13	Safety requirements for electronic control gear for use on AC supplies
IEC 61000-3-2	Electromagnetic compatibility (EMC) - Part 3-2: Limits for harmonics
IEEE 519-2014	Recommended Practice and Requirements for Harmonic Control
ES 7823	Energy efficiency requirements for electrical lamps
ES 7800	Tubular LED Lamps –designed-to retrofit fluorescent lamps-safety requirements
ES 7824	Tubular LED Lamps)-Performance requirement
IEEE 1789	Fliker. Standards & test methods
IEC 62504	Light emitting diode (LED) products and related equipment
IEC 62471	Safety of lamps and lamp system photo biological
IEC TR62778	The assessment of blue light hazard to light sources and luminaires
IEC 62612	Self-ballasted LED lamps for general lighting services with supply voltages > 50 V - Performance requirements

IEC 62560	Self-ballasted LED lamps for general lighting services with supply voltages > 50 V – safety specifications

3 – Environmental Conditions:

Ambient Temperature (°C)	-5 to45(50 as option)
Humidity	Max 90%.

4 – General Description

- LED tube light should be of a type of LED lamps (polycarbonate or glass according to ….EDC requirement) with G13 bases to replace traditional fluorescent tubes.
- As compared to fluorescent tubes, the most important advantages of LED tubes should be energy efficiency and long service life.
- LED tube lamp parts should be environmentally friendly (green designed)
- LED tubes should be typically installed in place of former fluorescent tubes in compatible luminaires as they are, with consideration of manufacturer and lamp-specific restrictions.
- The LED tubes should be non-flicker

5- Design criteria

The LED tubes of T8-9W, 60 cm (Milky) should be designed to withstand the minimum requirements of the following :

Voltage range	100-260VAC
Frequency (HZ)	50
Maximum power consumption	9 W+10% For each lamp and+7.5% average for all samples
Displacement factor	≥ 0.92 at rated voltage
THD	Not more than 20 % for Current
Total lumen	≥ 900 Lumen
Color temperature (CCT)	(3000 : 6500)K

	(to be defined by ….EDC) according to the application
Color rendering index (CRI)	≥ 80
Lamp life time	≥ 30000 Hr. (L80).
Base lamp (CAP)	G13
Tube dim	25mm
SDCM (standard deviation color matching)	≤5

6 - Inspection and Testing :

1) ..… EDC preserves its right to witness the routine testing of the lamps.
2) Unless otherwise specified or approved in writing by ……EDC, all lamps should be tested in accordance with all relevant latest applicable mentioned standards and test report certificate should be provided.
3) ….. EDC may carryout testing at its lab randomly. The lamp will be assumed as rejected if any functions of the lamp are found faulty

7- <u>Acceptance tests</u>

1. Aging test.
2. Insulation test using 500VDC voltage for 60 sec. the resistance shall be not less than 4 M .ohm
3. Insulation test using (2Uo + 1000) VAC voltage for 60 sec.
4. Mechanical Stress test.
5. Glow-wire test.
6. Electrical tests:
   i. Power
   ii. Displacement factor
   iii. THD
7. Optical tests:
   i. Lumen Flux
   ii. Efficacy
   iii. Color rendering

8. Life time tests:
   i. On/off
9. Operation time accelerating

6-Marking

The marking should be according to the following table

	Items	Place	
		Product *	Carton*
Marking	Mark of origin (trademark, manufactures name)	√	√
	Brand name	√	√
	Year of manufacturing	√	√
	Part number		√
	Patch Number		√
	Power (w)	√	√
	Voltage (V)	√	√
	Displacement factor	√	√
	Rated luminous flux (lm)	√	√
	Correlated Colour Temperature(CCT)	√	√
	Colour rendering index (CRI)		√
	Lamp photometric code		√
	Rated Life time		√
	Efficacy		√
	Model	√	√
	Energy efficiency card		√
	Base type	√	√
	SDCM		√

* The marks on the product and carton should not be erasable

7 – Packing

The lamps boxes should be packed in suitable cartoons to withstand transporting handling and storing

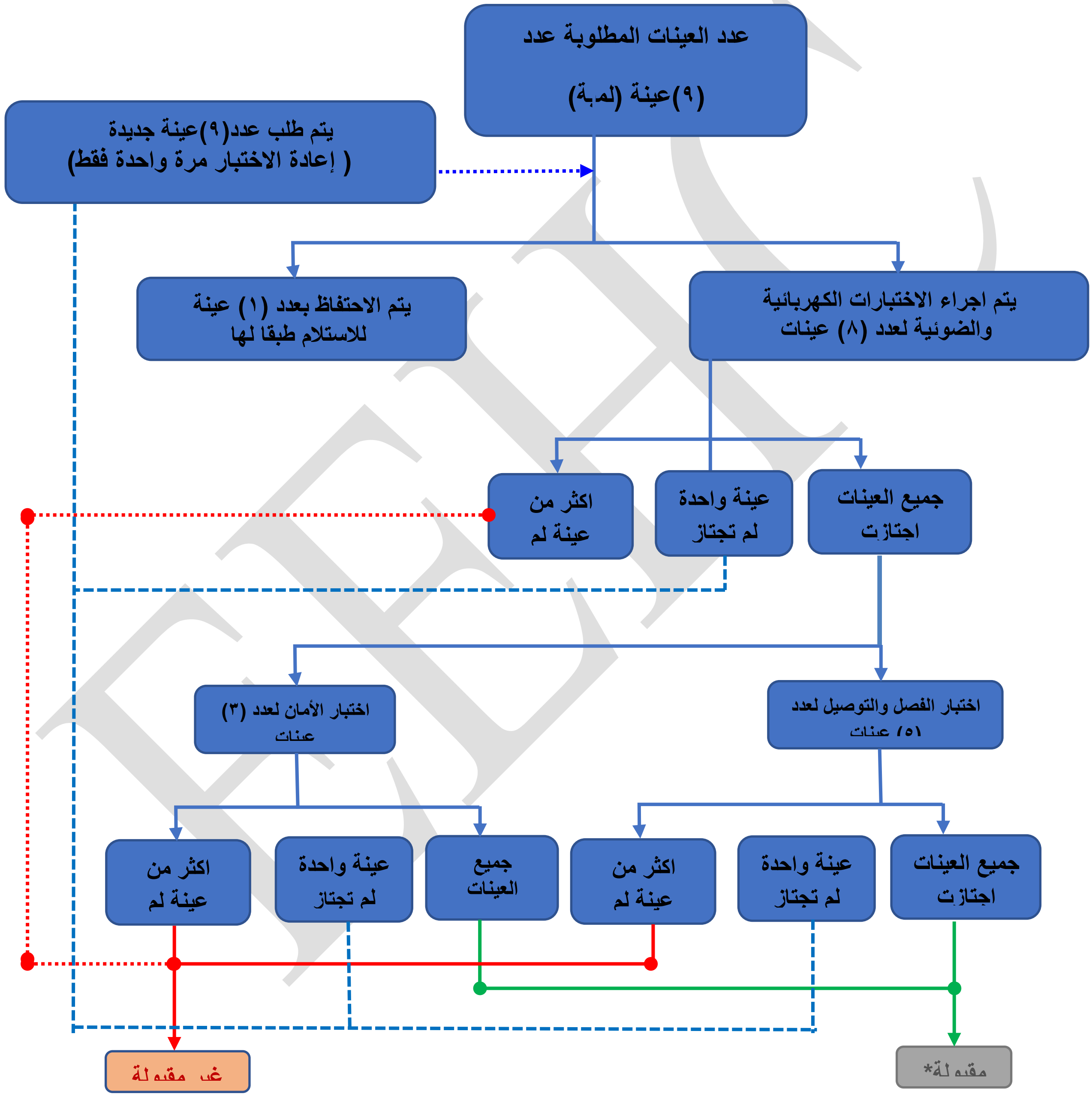
8 – Catalogues, samples and guarantee tables

- The manufacturer/tenderer should guarantee the lamp against all defects arising out of faulty design or workmanship three (3) years starting from delivery date..
- Factory test reports should be submitted.
- The tenderer should fill in thoroughly the attached guarantee technical data tables.
- The tenderer should attach with his offer the original catalogue for all components used with catalogue numbers marked clearly for the offered lamps (soft and hard copies).
- The tenderer should submit with his offer 9 random samples of each item (uncounted) 8 of them for testing, and the offer will be rejected wherever the test produce fails. (According to attached flow chart).

اشتراطات :-

1. يلزم الا تقل فترة الضمان للمبات الموردة عن ثلاث سنوات من تاريخ التوريد على ان تستبدل الكميات التالفة من اللمبات (بسبب عيوب التصميم او الصناعة وخلال فترة الضمان) خلال مدة لا تزيد عن أسبوع من تاريخ اخطار المورد بتلفها بلمبات أخرى جديدة ومن نفس الطراز والمواصفات هذا بالإضافة الى شهادة ضمان فني بعمر اللمبة.
2. يلزم تقديم شهادات الاختبارات النوعية من معامل معتمدة ومحايدة (ويفضل ان تكون المعامل مصرية والأولوية لمركز أبحاث الجهد الفائق).

مرفق تسلسل الأختبارات التي يتم أجراؤها علي العينات



*يلزم اجتياز جميع الاختبارات

9-Guarantee table

Designer company	
manufactures name	
Brand name	
Year of manufacturing	
Power	
Voltage range	
Frequency (HZ)	
Displacement factor	
THD	
Total lumen	
Color temperature (CCT)	
Color rendering index (CRI)	
Lamp life time	
Base lamp (CAP)	
Tube dim	
SDCM (standard deviation color matching)	
Model	
Base type	
Energy efficiency card	
Name of type test lab	
Guarantee technical period	